ELECTRONIC SUPPLEMENTARY INFORMATION (ESI)

Project AMBROSIUM v10.0: Technical & Mathematical Foundations

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Note S1: Electrodialytic Debye Screening Length Expansion Derivation

The Debye screening length λ_D in an electrolyte solution is given by:

$$\lambda_D = \sqrt{\frac{\varepsilon_r \varepsilon_0 k_B T}{2 N_A e^2 I}}$$

where I is the ionic strength. In physiological saline (150 mM NaCl), I = 150 mol/m³, yielding λ_D = 0.80 nm. Under active electrodialysis ion extraction, I drops to 3.86 mM, expanding λ_D to 4.98 nm (λ_D = 0.80 nm × sqrt(150 / 3.86) = 4.98 nm). The gate voltage is strictly clamped to $V_{gate} < 1.23$ V to prevent water electrolysis.

Note S2: 1D Thermal PDE Courant Stability Proof

The 1D transient heat conduction equation governing volumetric RF microwave absorption is:

$$\frac{\partial T}{\partial t} = \alpha_T \frac{\partial^2 T}{\partial x^2} + \frac{q_{rf}}{\rho c_p}$$

For an explicit finite difference scheme, the Courant-Friedrichs-Lewy (CFL) stability criterion mandates $r \leq 0.50$. For Δx = 0.5 µm and $\alpha_T = 1.43 \times 10^{-7}$ m²/s, the critical max time step is:

$$\Delta t_{CFL} = \frac{(\Delta x)^2}{2 \alpha_T} = 874.12 \text{ ns}$$

My operational selection of Δt = 100.0 ns yields r = 0.0572 << 0.50, guaranteeing unconditional numerical stability.

Note S3: Die-to-Flange Aerogel CTE Strain Buffering

The thermal expansion mismatch between the silicon substrate ($\alpha_{Si} = 2.6 \times 10^{-6}$ / K) and the copper waveguide flange ($\alpha_{Cu} = 16.5 \times 10^{-6}$ / K) creates interface shear stress during 72 °C RF gelation. Flanking the cavity with silica aerogel gaskets (k = 0.015 W/(m·K)) attenuates thermal gradients and maintains structural integrity.

Note S4: Human-in-the-Loop Methodology & Author Declaration

I personally executed all physical derivations, hardware HDL programming, simulation modeling, and metrology audits for Project AMBROSIUM v10.0. Gemini Notebook was utilized strictly for LaTeX compilation and document layout structuring.